

Dataset Description & Metadata: NeuroVision

- **Project Title:** NeuroVision - Brain Tumor Classification
- **Core Objective:** To classify human brain Magnetic Resonance Imaging (MRI) scans into '**Tumor**' (**Yes**) or '**Normal**' (**No**) categories using a deep learning Convolutional Neural Network (CNN).

1. Dataset Source & Origin

- **Origin:** The dataset consists of brain MRI scans sourced from open-source repositories like Kaggle.
- **File Structure:** The original raw data was provided in a compressed .zip archive containing separate, nested folders for each target class (yes and no).

2. Dataset Specifications & Class Distribution

- **Total Number of Images:** 253 MRI scans
- **Number of Classes:** 2 Classes (Binary Classification Problem)
 - **Class 0 (Normal / 'no'):** Healthy brain MRI scans containing no visible tumor masses or structural deformations.
 - **Class 1 (Tumor / 'yes'):** Brain MRI scans clearly showing abnormal white patches, tissue swelling, or structural deformations indicative of a tumor.

3. Preprocessing & Splitting Specifications

Before the images were passed into the CNN architecture, the following data pipeline was applied:

- **Color Conversion:** Converted from multi-channel color spaces to **Grayscale** (single channel) to minimize computational overhead while retaining vital structural details.
- **Dimensionality Reduction (Resizing):** All scans were uniformly resized to a fixed resolution of **128 x 128 pixels** to provide consistent input shapes for the neural network.
- **Min-Max Normalization:** Raw pixel intensity values ranging from 0 to 255 were divided by 255.0 to scale them down to a **0.0 to 1.0 range**. This step prevents exploding/vanishing gradient issues and allows the Adam optimizer to converge much faster.
- **Train-Test Data Split:** The total 253 images were randomly shuffled and split using an **80:20 ratio**:

- **Training Set:** 202 images utilized for training model parameters and weights.
- **Testing Set:** 51 unseen images held back to rigorously evaluate actual model performance.